JAVA TECHNICAL TRAINING PRODUCT BATCH-I
SESSION:03 DATE:24/02/2026

HASHING

finding common elements in an array
(order of second array)

```
import java.util.*;
public class A6_O {
   public static void main(String[] args)
   {
       int[] arr1={10,1,3,5,7,9};
       int[] arr2={2,5,3,12,13,14};
       int[] hash= new int[102];
       int i;
       for( i=0;i<arr1.length;i++)
       {
           hash[arr1[i]]=1;
       }
       for(i=0;i<arr2.length;i++)
       {
           if(hash[arr2[i]]==1)
           {
               System.out.print(arr2[i]+" ");
           }
       }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-
5 3
Process finished with exit code 0
```

.

(order of first array)

```
import java.util.*;
public class A6_O {
   public static void main(String[] args)
   {
       int[] arr1={10,1,3,5,7,9};
       int[] arr2={2,5,3,12,13,14};
       int[] hash= new int[102];
       int i;
       for( i=0;i<arr2.length;i++)
       {
           hash[arr2[i]]=1;
       }
       for(i=0;i<arr1.length;i++)
       {
           if(hash[arr1[i]]==1)
           {
               System.out.print(arr1[i]+" ");
           }
       }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\
3 5
Process finished with exit code 0
```

(sorted order)

```
import java.util.*;
public class A6_O {
   public static void main(String[] args)
   {
       int[] arr1={10,1,3,5,7,9};
       int[] arr2={2,5,3,12,13,14};
       Arrays.sort(arr1);
       int[] hash= new int[102];
       int i;
       for( i=0;i<arr2.length;i++)
       {
           hash[arr2[i]]=1;
       }
       for(i=0;i<arr1.length;i++)
```

```
        {
            if(hash[arr1[i]]==1)
            {
                System.out.print(arr1[i]+" ");
            }
        }
    }
}
```

Run A6_O

```
C:\Users\cse&student\.jdks\openjdk-25.0.2\
3 5
Process finished with exit code 0
```

(Handling duplicates)

```
import java.util.*;
public class A6_O {
   public static void main(String[] args)
   {
       int[] arr1={10,1,3,3,3,3,5,7,9};
       int[] arr2={2,5,3,3,12,13,14};
       Arrays.sort(arr1);
       int[] hash= new int[102];
       int i;
       for( i=0;i<arr2.length;i++)
       {
           hash[arr2[i]]=1;
       }
       for(i=0;i<arr1.length;i++)
       {
           if(hash[arr1[i]]==1)
           {
               System.out.print(arr1[i]+" ");
              hash[arr1[i]]=0;
           }
       }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\
3 5
Process finished with exit code 0
```

JAVA COLLECTIONS

Array List

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(200);
       l1.add(300);
       l1.add(400);
       System.out.println("Size:"+l1.size());
       System.out.println("0th element:"+l1.get(0));
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\Intelli
Size:4
0th element:100
Element:100
Element:200
Element:300
Element:400

Process finished with exit code 0
```

Removing and updating element

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(200);
       l1.add(300);
       l1.add(400);
       System.out.println("Size:"+l1.size());
       System.out.println("0th element:"+l1.get(0));
       l1.remove(0); //removed 100
       l1.set(0,2000);//modifying the element 200 to 2000
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }

   }
}
```

```
Run    A6_O

C:\Users\cse&student\.jdks\openjdk-25.0.2\b
Size:4
0th element:100
Element:2000
Element:300
Element:400

Process finished with exit code 0
```

Sorting

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(200);
       l1.add(500);
       l1.add(300);
       l1.add(400);
       System.out.println("Size:"+l1.size());
       System.out.println("0th element:"+l1.get(0));
       l1.remove(0); //removed 100
       l1.set(0,2000);//modifying the element 200 to 2000
       Collections.sort(l1);
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }
```

```
    }
}
```

```
Run    A6_O

C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaage
Size:5
0th element:100
Element:300
Element:400
Element:500
Element:2000

Process finished with exit code 0
```

Removing particular element

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(200);
       l1.add(500);
       l1.add(300);
       l1.add(400);
       System.out.println("Size:"+l1.size());
       System.out.println("0th element:"+l1.get(0));
       l1.remove(Integer.valueOf(100)); //removed 100 by value not indexing
       l1.set(0,2000);//modifying the element 200 to 2000
       Collections.sort(l1);
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }

   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Progr
Size:5
0th element:100
Element:300
Element:400
Element:500
Element:2000

Process finished with exit code 0
```

Removing multiple elements

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(100);
       l1.add(500);
       l1.add(300);
       l1.add(400);
       l1.add(100);
       System.out.println("Size:"+l1.size());
       System.out.println("0th element:"+l1.get(0));
       l1.removeIf(n->n==100); //removed 100 by value not indexing
//        l1.set(0,2000);//modifying the element 200 to 2000
//        Collections.sort(l1);
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }

   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.
Size:6
0th element:100
Element:500
Element:300
Element:400

Process finished with exit code 0
```

Reversing elements

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(100);
       l1.add(500);
       l1.add(300);
       l1.add(400);
       l1.add(100);
       System.out.println("Size:"+l1.size());
//        System.out.println("0th element:"+l1.get(0));
//        l1.removeIf(n->n==100); //removed 100 by value not indexing
//        l1.removeAll(Arrays.asList(100));
       l1.set(0,2000);
       Collections.sort(l1);
      Collections.reverse(l1);
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }

   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\b
Size:6
Element:2000
Element:500
Element:400
Element:300
Element:100
Element:100

Process finished with exit code 0
```

PARTIAL SORTING

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
      List<Integer> l1=new ArrayList<>();
      l1.add(100);
      l1.add(200);
       l1.add(500);
       l1.add(300);
       l1.add(400);
       l1.add(10);
       System.out.println("Size:"+l1.size());
//        System.out.println("0th element:"+l1.get(0));
//        l1.removeIf(n->n==100); //removed 100 by value not indexing
//        l1.removeAll(Arrays.asList(100));
//       l1.set(0,2000);
       Collections.sort(l1);
      Collections.reverse(l1.subList(0,4));
       for(i=0;i<l1.size();i++)
       {
           System.out.println("Element:"+l1.get(i));
       }

   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagen
Size:6
Element:300
Element:200
Element:100
Element:10
Element:400
Element:500

Process finished with exit code 0
```

HASH MAP

Declaration,storing key value pairs , accessing and checking for key

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int arr[]={1,2,10,12,13};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for(int i=0;i<arr.length;i++) {
           h1.put(arr[i], 1);
       }
          if( h1.containsKey(10))
          {
              System.out.println(10);
          }

   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin
10

Process finished with exit code 0
```

Finding common elements between 2 arrays using HashMap

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int arr1[]={1,2,3,4,5};
       int arr2[]={1,2,10,12,13};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for(int i=0;i<arr1.length;i++)
       {
           h1.put(arr1[i],1);
       }
       for(int i=0;i<arr2.length;i++) {
           if (h1.containsKey(arr2[i])) {
               System.out.print(arr2[i] + " ");
           }
       }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\b
1 2
Process finished with exit code 0
```

(to handle duplicates)

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int arr1[]={1,1,2,2,3,4,5};
       int arr2[]={1,2,10,12,13};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for(int i=0;i<arr1.length;i++)
       {
           h1.put(arr1[i],1);
       }
       for(int i=0;i<arr2.length;i++) {
           var value=h1.get(arr2[i]);
           if (value!=null && value==1) {

                   System.out.print(arr2[i] + " ");
                   h1.put(arr2[i],2);
               }
           }
       }
   }
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\j
1 2
Process finished with exit code 0
```

(handle duplicates **without exception handling for LEARNING PURPOSES**)

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int arr1[]={1,1,2,2,3,4,5};
       int arr2[]={1,2,10,12,13};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for(int i=0;i<arr1.length;i++)
       {
           h1.put(arr1[i],1);
       }
       for(int i=0;i<arr2.length;i++) {
           int value=h1.get(arr2[i]);
           if ( value==1) {

                   System.out.print(arr2[i] + " ");
                   h1.put(arr2[i],2);
               }
           }
       }
   }
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\In
1 2 Exception in thread "main" java.lang.NullPointerException: Cannot invoke "java.lang.Integer.
    at A6_O.main(A6_O.java:13)

Process finished with exit code 1
```

//OUTPUT WITH EXCEPTION (TO BE LEARNED TO HANDLE WHILE CONSTRUCTING SOLUTIONS)

(using contains key with handling duplicates)

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int arr1[]={1,1,2,2,3,4,5};
       int arr2[]={1,2,10,12,13};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for(int i=0;i<arr1.length;i++)
       {
           h1.put(arr1[i],1);
       }
       for(int i=0;i<arr2.length;i++) {
//             var value=h1.get(arr2[i]);
//             if (value!=null && value==1) {
//
//
//                 System.out.print(arr2[i] + " ");
//                 h1.put(arr2[i],2);
//             }
           if(h1.containsKey(arr2[i]))
           {
               System.out.println(arr2[i]);
               h1.remove(arr2[i]);
           }
       }
   }
}
```

```
Run    A6_O

C:\Users\cse&student\.jdks\openjdk-25.0.2\b
1
2

Process finished with exit code 0
```

Frequency of maximum element using hash map

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
       int arr[]={1,2,1,2,3,3,3};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for( i=0;i<arr.length;i++)
       {
          var it=h1.get(arr[i]);
          if( it!=null)
          {
              h1.put(arr[i],it+1);
          }
          else
          {
              h1.put(arr[i],1);
          }
       }
       System.out.println("Freq:"+h1.get(3));//frequency of max element in the
array


   }
}
```

```
Run    A6_O ×

C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-ja
Freq:3

Process finished with exit code 0
```

(Displaying every frequency pair (hashmap))

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
```

```java
        int i;
        int arr[]={1,2,1,2,3,3,3};
        HashMap<Integer,Integer> h1=new HashMap<>();
        for( i=0;i<arr.length;i++)
        {
           var it=h1.get(arr[i]);
           if( it!=null)
           {
               h1.put(arr[i],it+1);
           }
           else
           {
               h1.put(arr[i],1);
           }
        }
        //System.out.println("Freq:"+h1.get(3));//frequency of max element in
the array
        for(var it:h1.entrySet())
        {
            int key=it.getKey();
            int value=it.getValue();
            System.out.println(key+" "+value);
        }

    }
}
```

Run A6_O

```
C:\Users\cse&student\.jdks\openjdk-25.0.2\b
1 2
2 2
3 3

Process finished with exit code 0
```

Practice: most frequent element in an array

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
       int arr[]={1,2,1,2,3,3,3};
       HashMap<Integer,Integer> h1=new HashMap<>();
       for( i=0;i<arr.length;i++)
       {
          var it=h1.get(arr[i]);
          if( it!=null)
          {
              h1.put(arr[i],it+1);
          }
          else
          {
              h1.put(arr[i],1);
          }
       }
       //System.out.println("Freq:"+h1.get(3));//frequency of max element in
the array
       int max=-1, maxkey=-1;
       for(var it:h1.entrySet())
       {
           int key=it.getKey();
           int value=it.getValue();
           if(max<value) {
               max = value;
               maxkey= key;
           }

       }
       System.out.println("MOSTFREQUENT ELEMENT:"+maxkey);
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Program
MOSTFREQUENT ELEMENT:3

Process finished with exit code 0
```

Remove duplicates

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
       List<Integer> l1=new ArrayList<>();
       HashMap<Integer,Integer> h1=new HashMap<>();
       l1.add(1);
       l1.add(2);
       l1.add(3);
       l1.add(4);
       l1.add(1);
       l1.add(2);
       List<Integer> res=new ArrayList<>();
       for( i=0;i<l1.size();i++)
       {
           if(h1.containsKey(l1.get(i))) {
               l1.remove(i);
           }
           else {
               h1.put(l1.get(i), 1);
               res.add(l1.get(i));
           }
       }
      for(var it:res)
      {
          System.out.print(it+":");
      }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java
1:2:3:4:
Process finished with exit code 0
```

Method 2

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
       List<Integer> l1=new ArrayList<>();
       HashMap<Integer,Integer> h1=new HashMap<>();
       l1.add(1);
       l1.add(2);
       l1.add(3);
       l1.add(4);
       l1.add(1);
       l1.add(2);
//        List<Integer> res=new ArrayList<>();
       for( i=0;i<l1.size();i++)
       {
           if(h1.containsKey(l1.get(i))) {
               l1.remove(i);
               i=i-1;
           }
           else {
               h1.put(l1.get(i), 1);
//                res.add(l1.get(i));
           }
       }
      for(var it:l1)
      {
          System.out.print(it+":");
      }
   }
}
```

```
Run    A6_O
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-jav
1:2:3:4:
Process finished with exit code 0
```

STRINGS

Declaration, accessing each character , displaying the size

```java
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
      String s1="sree sai ram";
      s1=s1+"AGH";
      for(int i=0;i<s1.length();i++)
      {
          System.out.println(s1.charAt(i));
      }
       System.out.println(s1.length());
   }
}
```

```
Run    A6_O

C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaagent:C:\Pr
s
r
e
e

s
a
i

r
a
m
A
G
H
15

Process finished with exit code 0
```

Frequency of characters in a string

```java
import java.util.*;//includes collections in java
public class A {
   public static void main(String[] args)
   {
     String s1="ansa snjsa snjs ks ksk s";
     HashMap<Character,Integer> h1=new HashMap<>();
     for(int i=0;i<s1.length();i++)
     {
         h1.put(s1.charAt(i),h1.getOrDefault(s1.charAt(i),0)+1);
     }
       for(var it:h1.entrySet()) {
           char key = it.getKey();
           int value = it.getValue();
           System.out.println("char:"+key+"-"+"value:"+value);
       }

   }
}
```

Run A

```
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin
char: -value:5
char:a-value:3
char:s-value:8
char:j-value:2
char:k-value:3
char:n-value:3

Process finished with exit code 0
```

Brute force Approach

Java (21)

Start Timer

```java
class Solution {
    public int maxSubarraySum(int[] arr, int k) {
        int sum=0,max=0,i,j;

        for(i=0;i<arr.length;i++)
        {
            j=i;
            sum=0;
            int tempk=k;
            while(j<arr.length && tempk!=0)
            {
                sum=sum+arr[j];
                j++;
                tempk--;
            }
            max=Math.max(max,sum);
        }
    return max;

    }
}
```

Output Window

Compilation Results | Custom Input

Compilation Completed

• Case 1

Input:

arr[] =

100 200 300 400

k =

2

Your Output:

700

OPTIMAL APPROACH

Java (21) | Start Timer

```java
class Solution {
    public int maxSubarraySum(int[] arr, int k) {
        int sum=0,max,i;

        for(i=0;i<k;i++)
        {
            sum+=arr[i];
        }
        max=sum;
            for(i=k;i<arr.length;i++)
            {
                sum=sum+arr[i]-arr[i-k];
                max=Math.max(max,sum);
            }

    return max;

    }
}
```

Output Window

Compilation Results | Custom Input | Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed	Attempts : Correct / Total
1111 / 1111	**1 / 1**
	Accuracy : 100%

Points Scored	Time Taken
2 / 2	**0.47**
Your Total Score: 2 ↑	

HASH SET
DECLARATION , CHECKING ELEMENTS

```
import java.util.*;//includes collections in java
public class A6_O {
   public static void main(String[] args)
   {
       int i;
       Set<Integer> h1=new HashSet<>();
       h1.add(10);
       h1.add(20);
       h1.add(30);
       h1.add(20);
       h1.add(10);
       h1.add(10);
       if(h1.contains(100))
       {
           System.out.println("Value is present");
       }
       else
       {
           System.out.println("Value is not present");
       }

   }
```

```
}
```

Run A6_O

```
C:\Users\cse&student\.jdks\openjdk-25.0.2\bin\java.exe "-javaage
Value is not present

Process finished with exit code 0
```